The Chinese University of Hong Kong, Shenzhen
School of Data Science

MAT3007 · Homework 6

Due: 11:59 pm, July 21, 2023

Instructions:

- Homework problems must be carefully and clearly answered to receive full credit. Complete sentences that establish a clear logical progression are highly recommended.
- You must submit your assignment in Blackboard. If some problems need coding, you should paste your code in the file. Please upload only **one** file(pdf). The file name should be in the format **last name-first name-hw6**.
- The homework must be written in English.
- Late submission will not be graded.
- Each student **must not copy** homework solutions from another student or from any other source.

Problem 1 (20pts). Suppose that $f : \mathbf{R} \to \mathbf{R}$ is convex, and $a, b \in \operatorname{dom} f$ with $a < b$, where dom denotes the domain of the function. More specifically, $f : \mathbf{R}^p \to \mathbf{R}^q$ means that f is an $\mathbf{R}^p$-valued function on some *subset* of $\mathbf{R}^p$, and this *subset* of $\mathbf{R}^p$ is the domain of the function f. Show that

(a)
$$f(x) \leq \frac{b-x}{b-a} f(a) + \frac{x-a}{b-a} f(b), \text{ for all } x \in (a, b)$$

Hint: (Jensen's Inequality) If $p_1, ..., p_n$ are positive numbers which sum to 1 and f is a real continuous function that is convex, then

$$f\left(\sum_{i=1}^{n} p_i x_i\right) \leq \sum_{i=1}^{n} p_i f\left(x_i\right)$$

(b)
$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(x)}{b - x}$$

for all $x \in (a, b)$.

(c) Suppose f is differentiable. Use the result in (b) to show that:

$$f'(a) \leq \frac{f(b) - f(a)}{b - a} \leq f'(b)$$

Note that these inequalities also follow form:

$$f(b) \geq f(a) + f'(a)(b - a), \quad f(a) \geq f(b) + f'(b)(a - b)$$

(d) Suppose f is twice differentiable. Use the result in (c) to show that $f''(a) \geq 0$ and $f''(b) \geq 0$.

Solution:

(a) This is Jensen's inequality with $\lambda = (b-x)/(b-a)$.

(b) We obtain the first inequality by subtracting $f(a)$ from both sides of the inequality in (a). The second inequality follows from subtracting $f(b)$. Geometrically, the inequalities mean that the slope of the line segment between $(a, f(a))$ and $(b, f(b))$ is larger than the slope of the segment between $(a, f(a))$ and $(x, f(x))$, and smaller than the slope of the segment between $(x, f(x))$ and $(b, f(b))$.

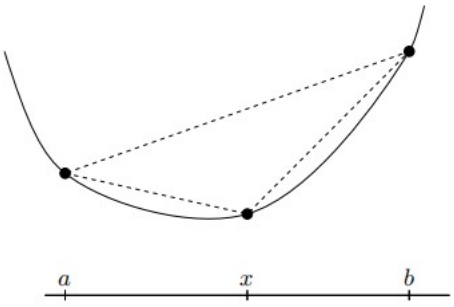


Figure 1: solution to ex1

(c) This follows from (b) by taking the limit for $x \to a$ on both sides of the first inequality, and by taking the limit for $x \to b$ on both sides of the second inequality.

(d) From part (c),

$$\frac{f'(b) - f'(a)}{b-a} \geq 0,$$

and taking the limit for $b \to a$ shows that $f''(a) \geq 0$.

Problem 2 (30pts) Show that the following functions are convex:

(a) $f(x) = -\log\left(-\log\left(\sum_{i=1}^m e^{a_i^T x + b_i}\right)\right)$ on $\operatorname{dom} f = \left\{x \mid \sum_{i=1}^m e^{a_i^T x + b_i} < 1\right\}$. You can use the fact that $\log\left(\sum_{i=1}^n e^{y_i}\right)$ is convex.

(b)

$$f(x,u,v) = -\log\left(uv - x^T x\right) \text{ on } \operatorname{dom} f = \left\{(x,u,v) \mid uv > x^T x, u, v > 0\right\}$$

(c) Let $T(x,\omega)$ denote the trigonometric polynomial

$$T(x,\omega) = x_1 + x_2 \cos\omega + x_3 \cos 2\omega + \cdots + x_n \cos(n-1)\omega$$

Show that the function

$$f(x) = -\int_0^{2\pi} \log T(x,\omega) d\omega$$

is convex on $\{x \in \mathbf{R}^n \mid T(x,\omega) > 0, 0 \leq \omega \leq 2\pi\}$.
Hint: Nonnegative weighted sum of convex functions is still convex. Let this property extend

to infinite sums and integrals. Assume that $f(x, y)$ is convex in x for each $y \in \mathcal{A}$ and $w(y) \geq 0$ for each $y \in \mathcal{A}$ and integral exists. Then the function g defined as

$$g(x) = \int_{\mathcal{A}} w(y) f(x, y) dy$$

is convex in x.

Solution:

(a) $g(x) = \log\left(\sum_{i=1}^{m} e^{a_i^T x + b_i}\right)$ is convex (composition of the log-sum-exp function and an affine mapping), so $-g$ is concave. The function $h(y) = -\log y$ is convex and decreasing. Therefore $f(x) = h(-g(x))$ is convex.

(b) We can express f as

$$f(x, u, v) = -\log u - \log\left(v - x^T x/u\right).$$

The first term is convex. The function $v - x^T x/u$ is concave because v is linear and $x^T x/u$ is convex on $\{(x, u) \mid u > 0\}$. Therefore the second term in f is convex: it is the composition of a convex decreasing function $-\log t$ and a concave function.

(c) The function

$$g(x, \omega) = -\log\left(x_1 + x_2 \cos\omega + x_3 \cos 2\omega + \cdots + + x_n \cos(n-1)\omega\right)$$

is convex in x for fixed ω. Therefore

$$f(x) = \int_0^{2\pi} g(x, \omega) d\omega$$

is convex in x.

Problem 3 (20pts). Consider the following function:

$$\begin{array}{ll} \text{minimize} & -x_1 - x_2 + \max\{x_3, x_4\} \\ \text{s.t.} & (x_1 - x_2)^2 + (x_3 + 2x_4)^4 \leq 5 \\ & x_1 + 2x_2 + x_3 + 2x_4 \leq 6 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{array}$$

(a) Verify this is a convex optimization problem.

(b) Use CVX to solve the problem.

Solution:

(a) The constraint $g_2(x) := x_1 + 2x_2 + x_3 + 2x_4 - 6$ is linear and thus, g_2 is convex. We continue with a discussion of $g_1(x) := (x_1 - x_2)^2 + (x_3 + 2x_4)^4 - 5$. The functions x^2 and x^4 are obviously convex and consequently, $g(x, y) = x^2 + y^4$ is also a convex mapping. We can now represent g_1 as follows:

$$g_1(x) = g(Ax), \quad \text{where } A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Hence, g_1 is convex since it is the composition of a convex and a linear function. Since the maximum and sum of convex functions is convex, the objective function $f(x) := -x_1 - x_2 + \max\{x_3, x_4\}$ is also convex.

```
n=4;
l=zeros(4,1);
cvx_begin
    variable x(n)
    minimize (-x(1)-x(2)+max(x(3),x(4)))
    subject to
    (x(1)-x(2))^2+(x(3)+2*x(4))^4 <=5;
    x(1)+2*x(2)+x(3)+2*x(4)<=6;
    x>=l;
cvx_end
```

Figure 2: solution to ex3

(b)

Problem 4 (15pts). To model the influence of price on customer purchase probability, the following logit model is often used:

$$\lambda(p) = \frac{e^{-p}}{1+e^{-p}}$$

where p is the price, $\lambda(p)$ is the purchase probability.
Assume the variable cost of the product is 0 (e.g., iPhone Apps). As the seller, you want to maximize the expected revenue by choosing the optimal price. That is, you want to solve:

$$\text{maximize}_p \quad p\lambda(p)$$

(a) Draw a picture of $r(p) = p\lambda(p)$ (for p from 0 to 10) and use the picture to show that $r(p)$ is not concave (thus maximize $r(p)$ is not a convex optimization problem)

(b) Write down p as a function of λ (the inverse function of $\lambda(p)$). Show that you can write the objective function as a function of $\lambda : \tilde{r}(\lambda)$, where $\tilde{r}(\lambda)$ is concave in λ.

(c) From part 2, write the KKT condition for the optimal λ. Then transform it back to an optimal condition in p.

Solution:

(a) See figure:

(b) Rewrite p as a function of λ.

$$\begin{aligned} & \lambda(p) = \tfrac{e^{-p}}{1+e^{-p}}, \\ \Longrightarrow \quad & p(\lambda) = \log(1-\lambda) - \log\lambda. \end{aligned}$$

Then the problem can be written as:

$$\text{maximize} \quad \lambda(\log(1-\lambda) - \log\lambda)$$

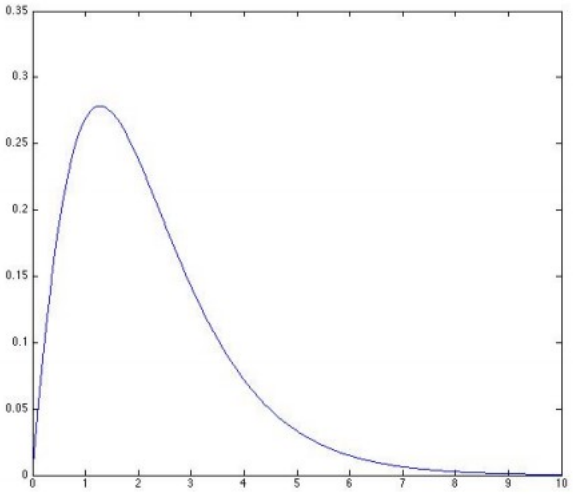


Figure 3: solution to ex4

Next, we show that this objective function is concave in λ. To see this, let $\tilde{r}(\lambda) = \lambda(\log(1-\lambda) - \log\lambda)$, take the derivatives, we have

$$\tilde{r}'(\lambda) = \log(1-\lambda) - \frac{\lambda}{1-\lambda} - \log\lambda - 1,$$
$$\tilde{r}''(\lambda) = -\frac{1}{1-\lambda} - \frac{1}{(1-\lambda)^2} - \frac{1}{\lambda} < 0.$$

The last inequality is because by definition, $0 < \lambda < 1$. Therefore, $\tilde{r}(\lambda)$ is concave. Maximizing $\tilde{r}(\lambda)$ is a convex problem.

(c) Optimal condition for λ,

$$\tilde{r}'(\lambda) = \log(1-\lambda^*) - \frac{\lambda^*}{1-\lambda^*} - \log\lambda^* - 1 = 0$$

Transform it back to optimal condition of p, we get

$$p^* - e^{-p^*} - 1 = 0$$

Problem 5 (15pts). Please use the definition of convex function or the condition of PSD Hessian matrix to derive the following first-order equivalence condition:

$$f(x) \geq f(y) + f^{'}(y)(x-y)$$

$\forall x, y \in \mathrm{dom} f$ if and only if f is a convex function.

Solution:

If f is a convex function we have $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$

$$\Rightarrow f(y + \lambda(x-y)) \leq \lambda f(x) + (1-\lambda)f(y) \text{ we set } \lambda > 0$$

we have $f(x) \geqslant f(y) + \frac{f(y+\lambda(x-y))-f(y)}{\lambda}$

$$\Rightarrow f(x) \geqslant f(y) + \frac{f(y+\lambda(x-y)-f(y)}{\lambda(x-y)}(x-y)$$

we take limit to 0 and hare $f(x) \geqslant f(y) + f^{'}(y)(x-y)$.

If we have $f(x) \geqslant f(y) + f'(y)\,(x-y)$, we set $z = \lambda x + (1-\lambda)y, \lambda \in [0,1]$, then we have:

$$f(x) \geqslant f(z) + f'(z)(x-z) \tag{1}$$

$$f(y) \geqslant f(z) + f'(z)(y-z) \tag{2}$$

$\lambda \cdot (1) + (1+\lambda) \cdot (2)$, we have

$$\lambda f(x) + (1-\lambda)f(y) \geq f(z) + f^{'}(z)[\lambda x + (1-\lambda)y - z] = f(z)$$

so we hare $\lambda f(x) + (1-\lambda)f(y) \geqslant f(z) = f(\lambda x + (1-\lambda)y)$, which completes the proof.